

Author Action Summary

The Face-Adjacency Graph of the Truncated Octahedron — Draft 9

Recommendation: conditional minor revision. No confirmed central mathematical error in the material reviewed. The previous originality/context and topology clarification requests have been answered.

Required before publication clearance

Independent complete tagged-release replay. Deposit the exact `face-graph-note-v9` state (commit, scripts, Lean sources, data and figure) in an immutable archive with identifier. Obtain a clean-environment independent execution of the full unoptimized Python verification runner and Lean 4/Mathlib v4.34.1 build; retain exit status, the `#print axioms` output and checksums of generated files. Source inspection alone does not close this requirement.

Recommended clarification (optional)

Make the distinction between Lean proof of a hand-entered matrix's characteristic polynomial and proof that the matrix represents the geometric graph/gauge especially transparent. The paper already lists what is not formalized; do not broaden those claims.

Changes satisfactorily incorporated

- Separate existing graph and previous monopole methods from the specific new magnetic factorizations.
- Include exact S^2 cycle-space ranks and a toroidal negative control.
- Add limited-scope Lean proof ledger (ordinary and special magnetic polynomials, part of K) and unverified portions.

Decision-reversal conditions

An independently verified matrix/gauge mismatch, a counterexample to the spherical proposition within its assumptions, or a substantive failure of the stated exact spectra would require renewed major review. An unexplained release-build failure delays clearance; it is not in itself proof of incorrect hand mathematics.